$$P(X=i;N) = \frac{C(r,i)\,C(N-r,n-i)}{C(N,n)}$$

$$P(X=i;N) > P(X=i;N-1)$$

$$\frac{\cancel{C(r,i)}\,C(N-r,n-i)}{C(N,n)} > \frac{\cancel{C(r,i)}\,C(N-1-r,n-i)}{C(N-1,n)}$$

$$\frac{\dfrac{\cancel{(N-r)!}\ N-r}{\cancel{(n-i)!}\,\cancel{(N-r-n+i)!}\ N-r-n+i}}{\dfrac{\cancel{N!}\ N}{\cancel{n!}\,\cancel{(N-n)!}\ N-n}} > \frac{\dfrac{\cancel{(N-r-1)!}}{\cancel{(n-i)!}\,\cancel{(N-r-n+i-1)!}}}{\dfrac{\cancel{(N-1)!}}{\cancel{n!}\,\cancel{(N-n-1)!}}}$$

$$\frac{N-r}{N-r-n+i} > \frac{N}{N-n}$$

$$\cancel{N^2} - \cancel{nN} - \cancel{rN} + rn > \cancel{N^2} - \cancel{Nr} - \cancel{Nn} + Ni$$

$$N < \frac{rn}{i}$$

$$P(X=i;N) > P(X=i;N+1)$$

$$\frac{nr}{i} > N > \frac{nr}{i+1}$$

x_1 objects out of n $\quad C(n, x_1)$

x_2 " " " $n-x_1$ $\quad C(n-x_1, x_2)$

x_3 " " " $n-x_1-x_2$ $\quad C(n-x_1-x_2, x_3)$

$\vdots$

x_k " " " $n-x_1-x_2-\cdots-x_{k-1}$ $\quad C(n-x_1-x_2-\cdots-x_{k-1}, x_k)$

$$\frac{n!}{x_1!\,\cancel{(n-x_1)!}} \; \frac{\cancel{(n-x_1)!}}{x_2!\,\cancel{(n-x_1-x_2)!}} \; \cdots \; \frac{(n-\cancel{x_1-x_2}-\cdots-x_{k-1})!}{x_k!\;0!}$$

Alternative form of multinomial

$$x_k = n - x_1 - x_2 - \cdots - x_{k-1}$$
$$p_k = 1 - p_1 - p_2 - \cdots - p_{k-1}$$

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_{k-1} = x_{k-1})$$
$$= \frac{n!}{x_1!\,x_2!\cdots(n-x_1-x_2-\cdots-x_{k-1})!}\; p_1^{x_1} p_2^{x_2} \cdots p_{k-1}^{x_{k-1}} (1-p_1-p_2\cdots-p_{k-1})^{x_k}$$